

Full Length Article

Joint entities and relations extraction method for complex scenarios of APT attack threat intelligence analysis

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ARTICLE INFO

Keywords:

Joint extraction
Overlapping triples
Threat intelligence analysis
APT

ABSTRACT

As cyber threat defense shifts from passive to proactive, the analysis of unstructured Cyber Threat Intelligence (CTI) has gained importance. This approach offers forward-looking insights into threat actors and attack techniques, especially in analyzing complex, multi-stage Advanced Persistent Threats (APTs). Entity and relation extraction helps uncover hidden associations between APT actors and their tactics via knowledge graphs. However, existing ontologies often have limited capability to adequately represent complex APT scenarios, especially in joint extraction tasks. Publicly available APT knowledge graph datasets are scarce, limiting model development. Moreover, APT-related CTI often involves chained narratives, resulting in overlapping triples that pose challenges for extraction models. To address these issues, we propose CAPTIO, an ontology tailored to APT intelligence, and construct CAPTIER, a corresponding dataset, which is publicly released. We define four types of overlapping triples and focus on three common in APT contexts. Based on these, we develop APT-OTJE, a joint extraction model using nested entity spans. This approach enhances feature extraction, task encoding, and relation pattern recognition compared to existing models. Experiments show that APT-OTJE achieves F1 improvements of 7% on CAPTIER and 25.2% on the HACKER dataset compared to baselines. Furthermore, our method is validated on two datasets for overlapping triple extraction. Case studies further compare extraction performance across baseline models for different triple categories.

1. Introduction

With the growing severity and complexity of cyber threats, attack methods have evolved from isolated techniques to multi-technology coordination supplemented by strategic tactics—most notably, Advanced Persistent Threats (APTs). Traditional passive defense and detection methods can no longer adequately address contemporary cybersecurity requirements. In response, proactive threat intelligence analysis has gained traction as a critical means to reduce the information asymmetry between attackers and defenders (Ren et al., 2023).

In particular, the rapid proliferation of threat intelligence text has created an urgent need for automated extraction of Cyber Threat Intelligence (CTI) from semi-structured or unstructured sources to facilitate timely knowledge base enhancement. Entity and relation extraction serves as a fundamental task in CTI analysis. It aims to identify key entities and their relationships within threat reports and transform them into structured triples (subject, relation, object) to enable knowledge

graph development.

APT attacks, with their multi-stage and chain-like nature, align well with the graph-based representation of knowledge. Consequently, research leveraging entity-relation extraction to address APT-specific challenges has been relatively active.

However, three critical challenges hinder APT-oriented entity-relation extraction research: **Q1**: Existing ontologies are insufficient for modeling the complexity of APT attack scenarios. Current joint extraction methods often define only a limited set of entity-relation types and constrain triples to fixed type patterns (e.g., head entity type, relation type, tail entity type), which oversimplifies real-world extraction scenarios. **Q2**: Although APT knowledge graph research has advanced in recent years, publicly available datasets remain scarce. The lack of standardized datasets hampers fair evaluation and comparison of proposed methods. **Q3**: APT attack paths often exhibit chain-like or divergent structures in knowledge graphs, making the issue of overlapping triples especially prominent. While the entity type “Attack Pattern (AP)”

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<https://doi.org/10.1016/j.cose.2026.104960>

Received 24 June 2025; Received in revised form 22 April 2026; Accepted 11 May 2026

Available online 12 May 2026

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is widely adopted across APT studies, overlapping triples involving this entity—particularly Subject-Object Overlap (SOO) and Nested Entity Overlap (NEO)—are rarely addressed in depth. Moreover, most existing joint extraction models exhibit limited effectiveness in handling such overlapping structures.

To tackle these challenges, this research builds upon recent advances in Natural Language Processing (NLP) to formally define overlapping triplets specific to the APT domain. Specifically, we categorize overlapping triples into four types, as shown in Fig. 1: Entity Pair Overlap (EPO), Single Entity Overlap (SEO), Subject-Object Overlap (SOO) and Nested Entity Overlap (NEO). NEO cases frequently occur when AP entities overlap with other types, particularly in mappings between APs and MITRE ATT&CK Tactics, Techniques, and Procedures (TTPs) identifiers. For instance, the phrase “identify recently logged-in users” corresponds to the attack technique “T1087”, where “logged-in users” represents a system “Identity” entity. Due to the broadly defined relation types in existing research (i.e., one relation may cover diverse semantic expressions), EPO occurs infrequently and is consequently not considered in the current analysis.

To address these challenges, this study extends existing APT attack ontologies to accommodate complex scenarios. Simultaneously, we constructed a dataset based on high-value ATT&CK cases. The inherent high information density and well-defined TTP relationships of these cases ensure dataset quality and accuracy. To effectively extract overlapping triples, we propose a span-based joint extraction model. By introducing task-specific feature extraction and encoding strategies, this approach strengthens triples extraction capabilities. Compared to existing joint extraction methods such as joint decoding (Liu et al., 2024), sequential traversal of entities (Shang et al., 2024), and sequence labeling (Ahmed et al., 2024; Zuo et al., 2022), our solution demonstrates greater specificity.

The contributions of this paper are as follows:

For Q1, we propose a comprehensive APT ontology, CAPTIO, which focuses on overlapping triple structures and complex semantic expressions, including fine-grained attack details and relation types with varying semantic directions. The ontology includes 13 entity types and 12 relation types. (Detailed in Chapter 3)

For Q2, based on CAPTIO, we manually label approximately 1500 instances to construct a high-quality APT knowledge graph dataset, CAPTIER, which comprises 7053 entities, 5907 relations, and 388 unique triple types. The dataset is publicly available at: <https://github.com/Bpig-C/CAPTIER>. (Discussed in Chapter 4.1.2)

For Q3, we propose APT-OTJE, a joint extraction method designed for overlapping triples in APT scenarios. It enhances the proactivity of shared feature representation during feature extraction compared to

prior work. Task encoding is improved through relative and regional position encoding to boost sub-task performance. And it employs a multi-head Biaffine mechanism to enhance the modeling of overlapping triples. (Proposed in Chapter 4, validated in Chapter 5)

The graphical abstract presents the research framework of this paper. To address three critical challenges in joint entity-relation extraction for complex APT scenarios, we propose a systematic solution. First, we define four overlapping triple types and develop the comprehensive CAPTIO ontology. Second, we construct the high-quality CAPTIER dataset as an evaluation benchmark. Finally, we design APT-OTJE, a joint extraction model tailored for overlapping scenarios. The effectiveness is validated through comparative experiments and ablation studies, with specialized experiments demonstrating our approach's superiority in overlapping triple extraction.

The remainder of this paper is structured as follows. Chapter 2 reviews related work on joint extraction and overlapping triple modeling. Chapter 3 introduces the proposed APT ontology, CAPTIO. Chapter 4 presents our span-based joint extraction approach. Chapter 5 describes experimental setup. Chapter 6 reports and discusses comparative experimental results. Chapter 7 discusses the robustness and limitations of current extraction models. Chapter 8 concludes the paper.

2. Related work

In recent years, research on Advanced Persistent Threats (APTs) has become increasingly active. Among various approaches, knowledge graph construction and extraction have emerged as critical techniques for CTI analysis. Entity-relation extraction in APT contexts is conventionally decomposed into two subtasks: Named Entity Recognition (NER) and Relation Extraction (RE), as explored in works such as APTNER (Wang et al., 2022) and HACKER (Luo et al., 2021).

Pipelined approaches achieve this by executing these two tasks sequentially. In the APT domain, research on named entity recognition has taken different focuses: comprehensive entity recognition across all threat intelligence components (Wang et al., 2022; Zhou et al., 2022), and specialized approaches targeting specific entity types such as tool entities (Liu et al., 2023).

However, these approaches suffer from error propagation between subtasks and lack effective information sharing, which limits their overall performance. To address these limitations, joint extraction methods have gained increasing attention. These methods simultaneously extract entities and relations, primarily using sequence labeling techniques (Ahmed et al., 2024; Zuo et al., 2022) or table-filling methods (Liu et al., 2024).

Nonetheless, these joint extraction studies generally face the

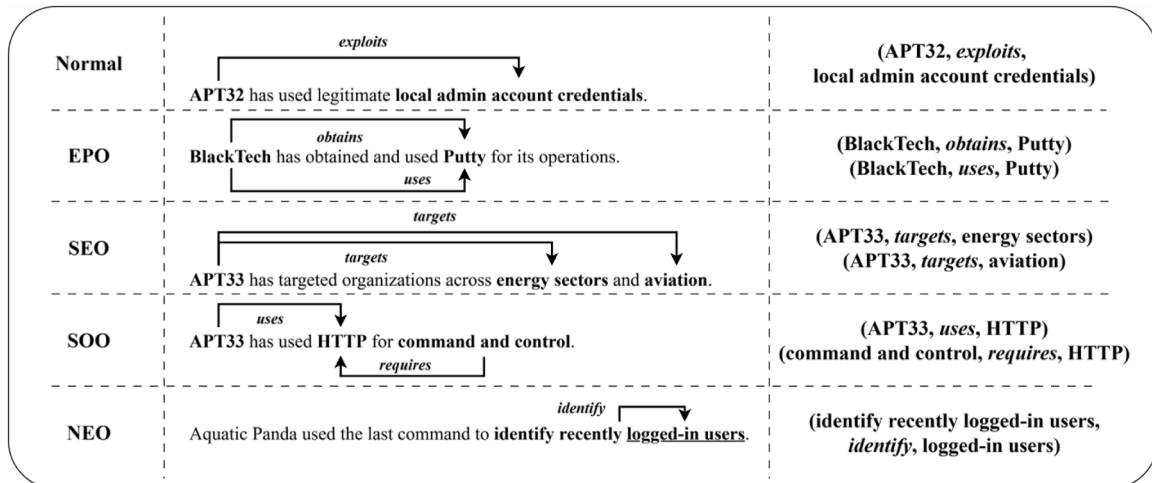


Fig. 1. Definition of overlapping triples in the APT domain.

challenge of overlapping triples. However, a systematic definition and experimental investigation specific to APT attacks has been lacking. Furthermore, some existing approaches are structurally constrained and cannot effectively resolve this issue. For example, sequence labeling-based joint extraction methods (Lv et al., 2024; Zuo et al., 2022) struggle with complex overlapping triples due to the uniqueness constraint in tag assignments. Work presented in Ahmed et al. (2024) builds upon Zuo et al. (2022) and introduces predefined relation matching, offering a preliminary solution for SEO and EPO problems. In this method, the number of sequence labels increases substantially as entity and relation types grow, leading to significantly degraded training performance.

Conversely, table-filling approaches Liu et al. (2024) can effectively reduce the redundancy inherent in complex type definitions and are capable of further extracting SOO overlaps. Some methods simplify complex tagging schemes by labeling only the head token to represent head/tail entities, relying on sophisticated joint decoding strategies to reconstruct entities and relations. Given the absence of such extraction methods in CTI, this paper references cutting-edge research from the medical field for comparison (Zhu et al., 2024). However, both sequence labeling and table-filling approaches struggle with position learning, limiting their ability to capture span-level semantic information for accurate relation classification.

Beyond sequence labeling and table-filling methods, only span-based joint extraction methods demonstrate effective capability in extracting all types of overlapping triples. Yet, in the current method applied to the Internet of Vehicles (IoV) domain (Shang et al., 2024), entity extraction relies on sliding window techniques to generate candidate spans, which introduce noise and reduce efficiency.

To overcome these limitations, our study proposes a joint extraction approach based on nested entity spans to APT scenarios. By enhancing feature extraction mechanisms and refining task encoding strategies, we improve both training efficiency and extraction accuracy, particularly in the presence of overlapping triples.

3. Complex APT intelligence ontology

To enable effective entity and relation extraction in complex APT scenarios, this study extends mainstream ontologies by refining the representation of attack process details and semantic associations, as shown in Table 1. Compared to prior research, our work achieves not only quantitative expansion in types but also enhances entity granularity by incorporating attack-phase details (e.g., Credentials, Observed Data, Configuration). Relations are semantically differentiated (e.g., uses, exploits, interacts-with) while introducing directional dependency semantics (e.g., passive expressions like requires). Additionally, this ontology supports bidirectional mapping with the STIX standard.

Notably, nested triples within this dataset predominantly involve APs and other entity types. This pattern occurs because attack techniques in APT scenarios often correspond to specific TTP identifiers defined in the MITRE ATT&CK framework, and these techniques frequently encapsulate or relate to multiple other entities and relations.

4. Method

In this section, we present the overall architecture of the proposed

Table 1
CAPTIO ontology definition.

Task	Type
Entity	Threat-Actor, Malware, Tool, Attack-Pattern, Vulnerability, Configuration, File, Infrastructure, Credential, Location, Identity, Industry, Observed-Data
Relation	targets, has, uses, exploits, interacts-with, requires, located-at, indicates, attributed-to, variant-of, affects, related-to

APT-OTJE (Overlapping Triple Joint Extraction for APT) model. The workflow of the model is shown in Fig. 2. It consists of the following components: a semantic embedding layer (SEL), a feature extraction layer (FEL), an entity position encoder (EPE), an entity span recognizer (ESR), a triple region encoder (TRE), and an overlapping triple extractor (OTE).

4.1. Target definition

In this paper, our objective is to extract entity-relation triples from APT intelligence texts. For an input text sequence from an APT report, denoted as $X = \{x_0, x_1, \dots, x_{N_{seq}-1}\}$, where N_{seq} is the number of tokens in the sequence, the model aims to output a set of triples $t \in T$, defined as:

$$T = \{(s, r, o) | s, o \in En, r \in Re\} \quad (1)$$

Here, En denotes the set of entities ($ent_1, ent_2, \dots, ent_{N_{ent}}$), where s is a subject entity and o is an object entity within En . Re represents the set of relation types ($rel_1, rel_2, \dots, rel_{N_{type}}$), with r signifying the relationship between s and o .

Ideally, the entity recognition task identifies all relevant entity spans within the text. Subsequently, the relation classification task determines the appropriate relation types linking each valid entity pair. To minimize training noise, this study employs ground-truth entities as input for the relation extraction component during the training phase.

4.2. Semantic embedding layer

This layer is based on the Transformer framework (Vaswani et al., 2017), utilizing the BERT series models (Devlin et al., 2019) to obtain contextualized embeddings for each token. These embeddings capture the deep semantic meanings of the tokens. To adapt to cybersecurity applications, we adopt the Plus version of SecureBERT (Aghaei et al., 2023), a pre-trained model fine-tuned on a large corpus of cybersecurity-related texts. This model is effective, particularly due to its use of domain-specific terminology commonly found in cybersecurity contexts. Specifically, the input is a token sequence segmented according to a predefined vocabulary, represented as $X = \{x_0, x_1, \dots, x_{N_{seq}-1}\}$.

$$E = \text{SecureBERT}(X) = \{e_0, e_1, \dots, e_{N_{seq}-1}\} \in \mathbb{R}^{N_{seq} \times d_e} \quad (2)$$

where d_e is the embedding dimension of the pre-trained model.

4.3. Feature extract layer

To address the limitations of traditional LSTM-based (Nguyen et al., 2021) hybrid feature representations, this study proposes an improved version of the Feature Partition Encoding (FPE) unit (Zhu et al., 2024). The original FPE model performs an accumulated softmax operation on the concatenated vectors at each time step within the LSTM network. It leverages the lower and higher index ranges of the hidden representation dimensions to construct entity gates and relation gates, respectively. The shared partition is statically constructed through element-wise multiplication of the entity and relation gates. Although this formulation accurately captures the shared features between entity and relation tasks from a mathematical perspective, it lacks adaptability and responsiveness. To address this limitation, we introduce Cross-Task Gating (CTG) in this section. CTG enhances the model's capability to actively perceive and filter cross-task shared partition features, enabling more dynamic and task-aware feature interaction, as shown in Fig. 3.

First, this module obtains the current timestep's entity partition ξ_{e_t} , relation partition ξ_{r_t} , and shared partition ξ_{s_t} using the original FPE paper's Eqs. (11)-(13):

$$\xi_{e_t}, \xi_{r_t}, \xi_{s_t} = \text{FPE}(e_t) \quad (3)$$

where e_t is the token embedding vector representation processed at the

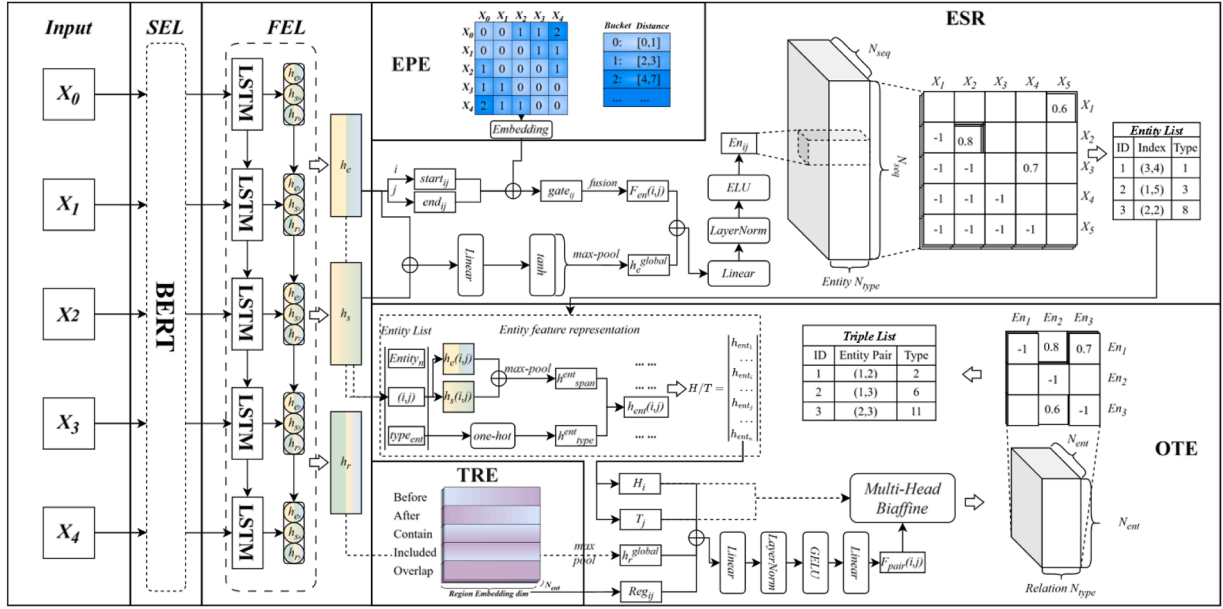


Fig. 2. Overall architecture of the APT-OTJE model.

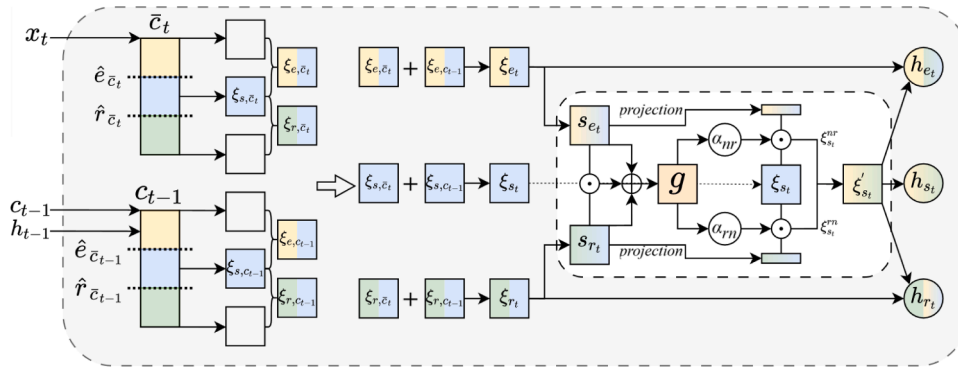


Fig. 3. Feature Extraction Module with CTG Integration.

current timestep. Subsequently, task state perception is performed using the entity and task partition features:

$$s_{e_t} = \text{GELU}(W_e \cdot \xi_{e_t} + b_e) \quad (4)$$

$$s_{r_t} = \text{GELU}(W_r \cdot \xi_{r_t} + b_r) \quad (5)$$

where $W_e, W_r \in \mathbb{R}^{d_l \times d}$ are learnable weight matrices, $d_l = \lfloor d/4 \rfloor$ is the dimension of the low-rank projection. $\text{GELU}(x) = x\Phi(x)$ is the Gaussian Error Linear Unit activation function. Next, a cross-task gating signal is constructed to learn the bidirectional contribution weights α_{nr} and α_{rn} between entity and relation tasks:

$$g = [s_{e_t} \oplus s_{r_t} \oplus (s_{e_t} \odot s_{r_t})] \quad (6)$$

$$\alpha_{nr} = \sigma(w_{nr}^\top g + b_{nr}) \quad (7)$$

$$\alpha_{rn} = \sigma(w_{rn}^\top g + b_{rn}) \quad (8)$$

where $\oplus$ denotes vector concatenation operation, $\odot$ denotes the element-wise multiplication operation, and $\sigma(z)$ represents the sigmoid activation function. Finally, the shared partition ξ'_{s_t} is dynamically reweighted via dual gating:

$$\xi_{s_t}^{ner-re} = \alpha_{nr} \cdot (\xi_{s_t} \odot P(s_{e_t})) \quad (9)$$

$$\xi_{s_t}^{ner-re} = \alpha_{nr} \cdot (\xi_{s_t} \odot P(s_{r_t})) \quad (10)$$

$$\xi'_{s_t} = \xi_{s_t}^{ner-re} + \xi_{s_t}^{ner-re} \quad (11)$$

where the projection function $P(v) = W_p \cdot v + b_p$, with $W_p \in \mathbb{R}^{d_l \times d}$, ensuring consistent dimensionality. The reconstructed shared partitions ξ'_{s_t} are then combined with task-specific features to obtain the final hidden state at the current time step:

$$h_{s_t} = \tanh(\xi'_{s_t}) \quad (12)$$

$$h_{e_t} = \tanh(\xi'_{s_t} + \xi_{e_t}) \quad (13)$$

$$h_{r_t} = \tanh(\xi'_{s_t} + \xi_{r_t}) \quad (14)$$

Finally, the representations are stacked over timesteps to obtain the partition representations for the entire input sequence $h_s, h_e, h_r \in \mathbb{R}^{N_{seq} \times d}$, where d is the hidden representation dimension.

4.4. Entity position encoder

This layer implements bucket-based relative position encoding tailored for the entity task. Due to the high similarity in embedding

vector representations of domain-specific nouns within vertical domains, relative position encodings are widely employed. However, the element-wise encoding paradigm of traditional methods significantly increases parameter volume, limiting training efficiency. Consequently, this section introduces an exponential bucketing scheme to preserve sensitivity to short entities while accommodating features of long entities.

Specifically, a base matrix D is first constructed based on the absolute distance between the i_{th} token and the j_{th} token in the input sequence: $D_{i,j} = |i - j|$, where $\forall i, j \in [0, N_{seq} - 1]$, $D \in \mathbb{N}^{N_{seq} \times N_{seq}}$. Then, the bucket-based discretization is applied to remap the base distance matrix:

$$B_{i,j} = \begin{cases} 0 & D_{i,j} \leq b_1 \\ N_{buc} & b_{N_{buc}} < D_{i,j} \leq b_{N_{buc}+1} \\ N_{BUC} & D_{i,j} > b_{N_{BUC}} \end{cases} \quad (15)$$

Where N_{BUC} is the total number of bucket boundaries, $N_{buc} \in \{1, 2, \dots, N_{BUC} - 1\}$ denotes an intermediate bucket index. $b_1, b_2, \dots, b_{N_{BUC}}$ are pre-defined bucket boundary values (logarithmically spaced in this study). Finally, the discrete bucket matrix embedding is used to obtain the bucketed relative position representation $Pos_{i,j}$ from token i to token j , which is then projected into the hidden dimension d :

$$Pos_{i,j} = \text{Embedding}(B_{i,j}) \in \mathbb{R}^d \quad (16)$$

4.5. Entity span recognizer

The objective of the entity recognizer is to identify all entity spans within the input sentence. To leverage the discriminative characteristics of different encoding layers, this paper first constructs a global feature representation h_e^{global} for entity tasks. Through the concatenation of entity task representations and shared representations, followed by linear transformation, tanh activation, and pooling, the current sequence's global feature representation is obtained as:

$$h_e^{global} = \text{maxpool}_{N_{seq}-1}(\tanh(\text{Linear}_{2d \rightarrow d}[h_s \oplus h_e])) \in \mathbb{R}^d \quad (17)$$

Subsequently, entity spans of size $N_{seq} \times N_{seq}$ are constructed using the input sequence length. Each element En_{ij} in the span matrix represents the concept representation sequence from $token_i$ to $token_j$ across various entity types. The vector representation of each element is formed by concatenating the corresponding start position feature $st_{ij} = h_e(i) \in \mathbb{R}^d$, end position feature $en_{ij} = h_e(j) \in \mathbb{R}^d$ and relative position encoding Pos_{ij} . Then, gating mechanisms are employed for feature fusion:

$$g_{eij} = \sigma(\text{Linear}_{3d \rightarrow d}[st_{ij} \oplus en_{ij} \oplus Pos_{ij}]) \in \mathbb{R}^d \quad (18)$$

$$F_{en}(i, j) = g_{eij} \odot st_{ij} + (1 - g_{eij}) \odot en_{ij} + Pos_{ij} \in \mathbb{R}^d \quad (19)$$

The prediction scores for each element's corresponding entity categories in the span matrix can then be obtained through concatenation and transformation of global feature representations and fusion feature representations:

$$ner_{ij} = \text{ELU}\left(\text{Dropout}\left(\text{LayerNorm}\left(\text{Linear}_{2d \rightarrow d}\left(\left[F_{en}(i, j) \oplus h_e^{global}\right]\right)\right)\right)\right) \in \mathbb{R}^d \quad (20)$$

$$En_{ij} = \sigma(\text{Linear}_{d \rightarrow N_{ent}^{type}}(ner_{ij})) \in \mathbb{R}^{N_{ent}^{type}} \quad (21)$$

Finally, this module extracts entity constituent lists $list_{en}$ from the original text using a threshold:

$$list_{en} = \{En_1^{type_{ent}}, En_2^{type_{ent}}, \dots, En_{N_{ent}}^{type_{ent}}\} \quad (22)$$

where $type_{ent}$ represents the corresponding entity type, and N_{ent} is the

number of entities extracted from the current sample.

4.6. Triple region encoder

After the model extracts all entity spans from the original text, it needs to further determine the categories of each pair of entities. The connectivity and directionality between entities significantly influence the results of category judgment. However, existing entity feature representations and global contextual representations cannot effectively address this issue. Therefore, this paper designs 5 types of inter-entity regional encoding categories: Before, After, Contain, Included, and Overlap. Specifically, the input consists of entity position spans $Pos_{en} \in \mathbb{Z}^{N_{ent} \times 2}$ that satisfy the requirements, where N_{ent} is the number of entities extracted by ESR, and 2 represents the entity's positive position indices. Here, the start index of the head entity is st_h , the end index is en_h , the start index of the tail entity is st_t , and the end index is en_t . The complete definitions of the 5 regional categories are as follows:

$$R_{type} = \begin{cases} B(h, t) = (en_h < st_t) \\ A(h, t) = (st_h > en_t) \\ C(h, t) = (st_h \leq st_t) \wedge (en_h > en_t) \wedge \neg B(h, t) \wedge \neg A(h, t) \\ I(h, t) = (st_t \leq st_h) \wedge (en_t > en_h) \wedge \neg B(h, t) \wedge \neg A(h, t) \\ O(h, t) = (st_h \leq st_t) \wedge (st_t \leq en_h) \wedge \neg C(h, t) \wedge \neg I(h, t) \wedge \neg B(h, t) \wedge \neg A(h, t) \end{cases} \quad (23)$$

The above five regional categories are then sequentially encoded, and their corresponding values are mapped to entity pairs as follows:

$$R_{ij} = \begin{cases} 0, & B(h, t) \\ 1, & A(h, t) \\ 2, & C(h, t) \\ 3, & I(h, t) \\ 4, & O(h, t) \end{cases} \quad (24)$$

Finally, the regional encoding between entity i and entity j is embedded into a unified hidden dimension d :

$$Reg_{ij} = \text{Embedding}(R_{ij}) \in \mathbb{R}^d \quad (25)$$

4.7. Overlap triple extractor

OTE identifies the relationship categories between entity spans based on features extracted by FEL and entity lists identified by ESR. This process specifically comprises three components: entity feature representation, Entity Pair Fusion Representation, and multi-head bidirectional attention representation.

4.7.1. Entity feature representation

First, this module needs to obtain the embedded representation of each entity span. These include entity span representations and type feature representations. Specifically, for the entity representation h_{span}^{ent} at token positions i and j :

$$h_{span}^{ent} = \frac{1}{j - i + 1} \sum_{n=i}^j [h_e(n) \oplus h_s(n)] \in \mathbb{R}^{2d} \quad (26)$$

where h_e and h_s are extracted from FEL, using entity cross-degree corresponding parts here. The type feature representation is obtained through one-hot encoding to represent the entity's corresponding type $type_{ent}$:

$$h_{type}^{ent} = \text{one}_{hot}(type_{ent}) \in \{0, 1\}^{N_{ent}^{type}} \quad (27)$$

where N_{ent}^{type} represents the total number of entity types. Finally, both are concatenated to form the complete representation of this entity:

$$h_{ent} = h_{span}^{ent} \oplus h_{type}^{ent} \in \mathbb{R}^{2d + N_{ent}^{type}} \quad (28)$$

4.7.2. Entity pair fusion representation

Subsequently, entity feature representations need to be combined to construct entity pair feature matrices. First, the head-tail entity list representation H/T is constructed:

$$H/T = \begin{bmatrix} h_{ent_1} \\ \dots \\ h_{N_{ent}} \end{bmatrix} \in \mathbb{R}^f \quad (29)$$

where the dimension $f = N_{ent} \times (2d + N_{rel}^{type})$, and N_{ent} represents the number of entities in the entity list. Then, the relational contextual representation is obtained through positional hidden vector averaging:

$$h_r^{global} = \frac{1}{N_{seq}} \sum_{i=1}^{N_{seq}} h_r(i) \in \mathbb{R}^d \quad (30)$$

where $h_r(i)$ represents the relational feature of the token at position i . Next, the head-tail entity span representations H_i/T_j , global relational contextual representation h_r^{global} , and triple regional encoding Reg_{ij} are fused. The fusion representation of each element in the entity pair feature matrix is $F_{pair}(i, j)$:

$$F_{pair}(i, j) = \begin{bmatrix} H_i \\ H_j \\ h_r^{global} \\ Reg_{ij} \end{bmatrix} \in \mathbb{R}^g \quad (31)$$

where the dimension $g = 2f + 2d$. Finally, multi-layer perceptrons are used to perform enhanced fusion and dimensionality reduction on entity pair representations:

$$\tilde{F}_{pair}(i, j) = \text{Linear}_{4d \rightarrow 2d}(\text{GELU}(\text{LayerNorm}(\text{Linear}_{g \rightarrow 4d}(F_{pair}(i, j)))))) \in \mathbb{R}^{2d} \quad (32)$$

4.7.3. Multi-head biaffine relation representation

Biaffine functions have been proven effective for inter-entity relation extraction. However, simple Biaffine approaches only learn single relational patterns with relatively low computational efficiency. Therefore, this paper introduces multi-head Biaffine attention to learning diverse relational patterns.

First, projections are performed on the head-tail entity span representations H_i/T_j :

$$\tilde{h}_i = \text{ReLU}(\text{Linear}_{f \rightarrow d}(H_i)) \in \mathbb{R}^d \quad (33)$$

$$\tilde{t}_j = \text{ReLU}(\text{Linear}_{f \rightarrow d}(T_j)) \in \mathbb{R}^d \quad (34)$$

Then, multi-dimensional reshaping is performed on the projections of the head-tail entity list representations using the number of heads k :

$$\hat{h}_i = \text{reshape}(\tilde{h}_i) \in \mathbb{R}^{k \times d_k} \quad (35)$$

$$\hat{t}_j = \text{reshape}(\tilde{t}_j) \in \mathbb{R}^{k \times d_k} \quad (36)$$

where the head dimension $d_k = d/k$. Then multi-head biaffine computation is performed:

$$B_{ij} = \sum_{n=1}^k (\hat{h}_i^{(n)})^T U_r^n \cdot \hat{t}_j^{(n)} \in \mathbb{R}^{N_{rel}^{type}} \quad (37)$$

where $\hat{h}_i^{(n)} \in \mathbb{R}^{d_k}$ represents the feature representation of the i_{th} head entity at the n_{th} head. Similarly, $\hat{t}_j^{(n)} \in \mathbb{R}^{d_k}$, and the relation parameter $U_r^n \in \mathbb{R}^{N_{rel}^{type} \times d_k \times d_k}$, where N_{rel}^{type} represents the total number of relation types. The computed relational matrix element representation is as follows:

$$S_{ij} = B_{ij} + W_r \tilde{F}_{pair}(i, j) + b_r \in \mathbb{R}^{N_{rel}^{type}} \quad (38)$$

where the relation weight matrix $W_r \in \mathbb{R}^{N_{rel}^{type} \times 2d}$ and bias $b_r \in \mathbb{R}^{N_{rel}^{type}}$. Finally, the sigmoid activation function is used to predict the probability distribution of entity pair (ent_i, ent_j) across various relation types:

$$P_r(ent_i, ent_j) = \sigma(S_{ij}) \quad (39)$$

4.8. Loss

The loss function during training consists of two parts: the named entity recognition module and the relation extraction module. Both entity and relation modules employ binary cross-entropy loss functions for computation. In this paper, they are denoted as $\mathcal{L}_{ner}$ and $\mathcal{L}_{rel}$, respectively. The formulas are as follows:

$$\mathcal{L}_{ner} = - \sum_{1 \leq i \leq j \leq N_{seq}} \mathbb{I}(m_{ij}^{(b)} = 1) [y_{ijk} \log(p_{ijk}) + (1 - y_{ijk}) \log(1 - p_{ijk})] \quad (40)$$

$$\mathcal{L}_{rel} = - \sum_{1 \leq i \leq j \leq N_{ent}} \mathbb{I}(m_{ij}^{(b)} = 1) [y_{ijk} \log(p_{ijk}) + (1 - y_{ijk}) \log(1 - p_{ijk})] \quad (41)$$

where $1 \leq b \leq \text{Batch}$, and $\mathbb{I}(m_{ij}^{(b)} = 1)$ indicates the presence of valid sequence and entity pair encodings. p_{ijk} represents the model's predicted entity and relation probabilities, while y_{ijk} represents the true entity and relation labels. Finally, the overall training loss is computed as the sum of both modules:

$$\mathcal{L} = \mathcal{L}_{ner} + \mathcal{L}_{rel} \quad (42)$$

5. Experimental setup

5.1. Baseline models

This experiment selects representative joint extraction models that are theoretically capable of extracting overlapping triples for comparison. These include models from the CTI domain (SMIEN) and non-CTI domains (SpERT, FPE). To ensure experimental accuracy, we apply the same experimental dataset to all models. However, SMIEN uses the paper data directly for comparison due to the uncertainties inherent in its syntactic dependency tree construction and its common use of the HACKER dataset with our approach. The baseline models are as follows:

SpERT (Eberts and Ulges, 2020): This method focuses on research in the NLP domain. It achieves end-to-end joint entity and relation extraction by enumerating text spans combined with local context classification and intra-sentence hard negative sampling.

FPE (Zhu et al., 2024): This method focuses on information mining from biomedical texts and designs a Feature Partition Encoding (FPE) framework. It effectively addresses the joint extraction problem of overlapping triples by dynamically partitioning neural features into entity, relation, and shared partitions.

SMIEN (Shang et al., 2024): This method proposes a Span-based Multi-variate Information-aware Embedding Network. It introduces perception modules for multi-dimensional information including semantic, temporal, dependency, and label information, along with their corresponding interaction mechanisms. It focuses on joint extraction problems in the threat intelligence domain where entities have high semantic similarity but weak associations.

5.2. Experimental datasets

Due to the extreme scarcity of publicly available annotated data for threat intelligence knowledge graphs, some studies have published extracted graphs but without corresponding original texts. Therefore, this paper constructs the CAPTIER dataset based on the CAPTIO ontology to validate the model's extraction performance on both overall and overlapping triples. Additionally, we utilize the open-source HACKER relation dataset for auxiliary validation.

5.2.1. CAPTIER dataset

We annotated 1500 intelligence instances from the ATT&CK based on our proposed data annotation framework (Qi et al., 2025) for experimentation and made them open-source. Since the original annotated data contains up to 388 unique triple types (head entity type, relation type, tail entity type), with the majority having extremely low frequencies, we selected 48 relatively high-frequency triple types to validate our method.

5.2.2. HACKER dataset

Luo et al. (2021) rapidly annotated a large number of relational triples through distant supervision. This dataset included over 9000 sentence instances and 22 different relation types from 2153 reports. After removing erroneous labels and consolidating entities and relations from identical sentences, we obtained 3593 sentence instances and 6071 relational triples.

5.2.3. Experimental metrics

(1) Baseline model experimental metrics.

To adequately reflect the performance across all categories of entity-relation problems, this paper adopts a Macro calculation method that first computes F1 scores and then takes their average:

$$Pre = \frac{1}{N_{ent}^{type}} \sum \frac{TP}{TP + FP} \quad (43)$$

$$Rec = \frac{1}{N_{ent}^{type}} \sum \frac{TP}{TP + FN} \quad (44)$$

$$F1 = \frac{1}{N_{ent}^{type}} \sum 2 \cdot \frac{Pre \times Rec}{P + R} \quad (45)$$

For the entity component, due to the excessive proportion of negative samples in the entity matrix caused by sequence length, only the metrics for correct samples are calculated. For relation classification, since the number of entities per sentence is relatively small, negative samples are introduced to enhance model convergence speed and training effectiveness.

(2) Overlapping triple experimental metrics.

To reflect the extraction performance of different types of overlapping triples, this paper adopts a Micro calculation method, where the overlapping type set is $C = \{SEO, SOO, NEO\}$. The calculation expression is as follows:

$$Pre_c = \frac{TP}{TP + FP} \quad (46)$$

$$Rec_c = \frac{TP}{TP + FN} \quad (47)$$

$$F1_c = 2 \cdot \frac{Pre_c \times Rec_c}{P + R} \quad (48)$$

Where the overlapping type $c \in C$. Since this metric is used solely for

model evaluation, negative samples are not introduced.

6. Experiments

6.1. Parameter settings

During the model construction process, we implemented all models using PyTorch. The experimental settings are shown in Table 2. The embedding dimensions and learning rates of each module are kept consistent. Due to the use of multi-class cross-entropy loss function, we select 0.5 as the threshold.

All experiments were conducted on a single NVIDIA V100 GPU. On the CAPTIER dataset, the average training time per epoch is approximately 20 s, and the full training process (150 epochs, covering both the NER and RE modules) requires approximately 1.7 h. On the HACKER dataset, despite its larger sample size, the lower triple complexity results in a shorter per-epoch training time. The full training process requires approximately 1.4 h. The average inference latency is approximately 4 ms per sentence.

6.2. Baseline model experiments

This experiment validates the performance comparison of current mainstream methods on both the CAPTIER dataset and the HACKER dataset, as shown in Table 3. On CAPTIER, although our entity-relation triple unique types (48 types) significantly exceed existing studies and include overlapping nested triples, the F1 score improved by 7% compared to previous methods.

On the HACKER relation dataset, we only conduct comparisons in the relation module. The experiments demonstrate that our method achieves a 25.2% improvement over existing studies. This performance gain stems from HACKER containing only 22 triple types and expressing only threat organization-related entities and relations, making extraction easier compared to the complex scenarios in CAPTIER.

6.3. Overlapping triple extraction experiments

To analyze the extraction performance of the trained model on different categories of overlapping triples, we evaluated the three types of overlapping triples on the test set, including validation data (Tables 4 and 5). The distribution statistics for both datasets are shown in Table 4. Triples between different types may overlap. SEO and SOO represent overlaps between two triples, while NEO represents nested overlap within a single triple.

The experimental results for three types of overlapping triples are shown in Table 5. During triple experiments, since baseline methods show low exact matching performance for overlapping triple extraction, we only conduct experiments using our proposed APT-OTJE model on both datasets. Based on the experimental analysis, SEO types are more easily identified due to their greater abundance. SOO types show the worst recognition performance because their patterns are more variable and diverse. The smaller data volume is likely the reason that limits NEO-type extraction performance. Additionally, the intermixing of multiple types of overlapping triples is a possible cause of the overall

Table 2
Model parameter settings.

Parameter	Value
Epoch	150
Hidden size	512
Batch size	16
Learning rate	2e-5
Dropout rate	0.2
Max sentence length	512
Max entity length	50
Threshold	0.5

Table 3

Performance of Models on CAPTIER and HACKER Datasets.

Model	CAPTIER						HACKER		
	NER			RE(Triple)			RE(Triple)		
	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
SpERT	0.740	0.722	0.730	0.483	0.391	0.432	0.416	0.431	0.423
FPE	0.722	0.705	0.713	0.489	0.349	0.407	0.419	0.352	0.383
SMIEN	-	-	-	0.471	0.412	0.442	0.459	0.435	0.447
Ours	0.771	0.764	0.763	0.623	0.466	0.516	0.777	0.651	0.699

Table 4

Quantity of different triplet types in CAPTIER and HACKER datasets.

Overlap Triple	CAPTIER			HACKER		
	Total	Trian	Test	Total	Trian	Test
SEO	4558	3515	1043	5948	4845	1103
SOO	1638	1307	331	302	237	65
NEO	427	334	93	11	9	2

poor extraction performance.

To further validate the impact of data volume on nested triple extraction performance, this paper employs up-sampling methods to enhance sentence instances containing NEO and conducts experiments on CAPTIER. Specifically, we adopt three up-sampling strategies: 1. Entity boundary word-level perturbation to simulate entity extraction result deviation; 2. Same-type entity replacement; 3. Supplementing nested triple associations within single sentences based on matching tables (the matching table is defined according to existing triple categories).

The experimental results in Table 6 demonstrate a positive correlation between data quantity and NEO entity performance. This confirms that adding relations based on existing high-frequency triple types is the most effective up-sampling method. However, when enhancing NEO triple sentences, the quantities of the other two triple types are inevitably increased as well, with SEO types generally outnumbering NEO types. Consequently, the experimental results show that SEO receives more obvious positive gains compared to NEO from this data volume increase.

Although the above experiments demonstrate the positive correlation between data volume and overlapping triple extraction performance, the model robustness issues brought by such up-sampling methods cannot be ignored. This paper further discusses and explains this in Chapter 7.

6.4. Ablation experiments

To thoroughly investigate the contribution of each module in our model structure to extraction performance, we conducted ablation experiments on the main modules of the model, as shown in Table 7. The experiments demonstrate that the word embedding model has a significant impact. Compared to the BERT model, SecureBERT improves entity performance by 4% and relation performance by 4.9%. The CTG in our feature extraction layer shows a more pronounced improvement in the entity module, likely because it actively introduces relation information into entity representation. Additionally, both task encoding modules EPE and TRE show significant effects.

Table 5

Extraction performance of Our APT-OTJE Method on different triplet types.

Datasets	SEO			SOO			NEO		
	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
HACKER	0.606	0.247	0.351	0.024	0.083	0.037	0.012	0.56	0.023
CAPTIER	0.389	0.185	0.251	0.092	0.105	0.098	0.136	0.201	0.162

6.5. Case study

To thoroughly investigate the model's extraction performance for different types of overlapping triples, this section conducts a case study. The case includes three sentences containing all three types of overlapping triples, compared with baseline models SpERT and FPE. The results are shown in Fig. 4, where green indicates correct predictions and red indicates incorrect predictions.

S1 represents typical SEO head entity overlap with a clear structure and relative frequency, which all three models can extract correctly. For S2, baseline models can correctly identify the first triple, but errors occur in subsequent triples in the long chain. SpERT correctly selects the head and tail entities but incorrectly determines the relation type. The FPE model shows the opposite pattern. For S3, SpERT correctly identifies triple types except NEO, but fails to accurately identify entities. FPE cannot effectively extract inter-triple relations due to a lack of entity semantic enhancement. Compared to baselines, our proposed APT-OTJE demonstrates more robust performance in SEO, SOO, and NEO overlapping scenarios, and can better extract entities and relations under certain conditions.

Table 6

NEO enhancement performance with three up-sampling methods.

Types	Boundary Perturbation	Same-type Replacement	Supplemental Triples
SEO	+6.77%	+1.95%	+7.34%
SOO	+2.21%	+1.07%	+2.85%
NEO	+6.42%	+4.86%	+6.75%

Note: '+' indicates the F1-score improvement after up-sampling on the corresponding overlapping triple type.

Table 7

Ablation experiments of each module in the APT-OTJE model.

Model	NER			RE		
	Pre.	Rec.	F1	Pre.	Rec.	F1
w/o SecureBERT	0.737	0.723	0.723	0.572	0.395	0.467
w/o CTG	0.743	0.759	0.747	0.614	0.449	0.492
w/o EPE	0.746	0.763	0.749	-	-	-
w/o TRE	-	-	-	0.578	0.434	0.480
1H-Biaffine	-	-	-	0.614	0.448	0.499
64H-Biaffine	-	-	-	0.629	0.437	0.500
256H-Biaffine	-	-	-	0.604	0.453	0.502
Ours(128H)	0.771	0.764	0.763	0.623	0.466	0.516

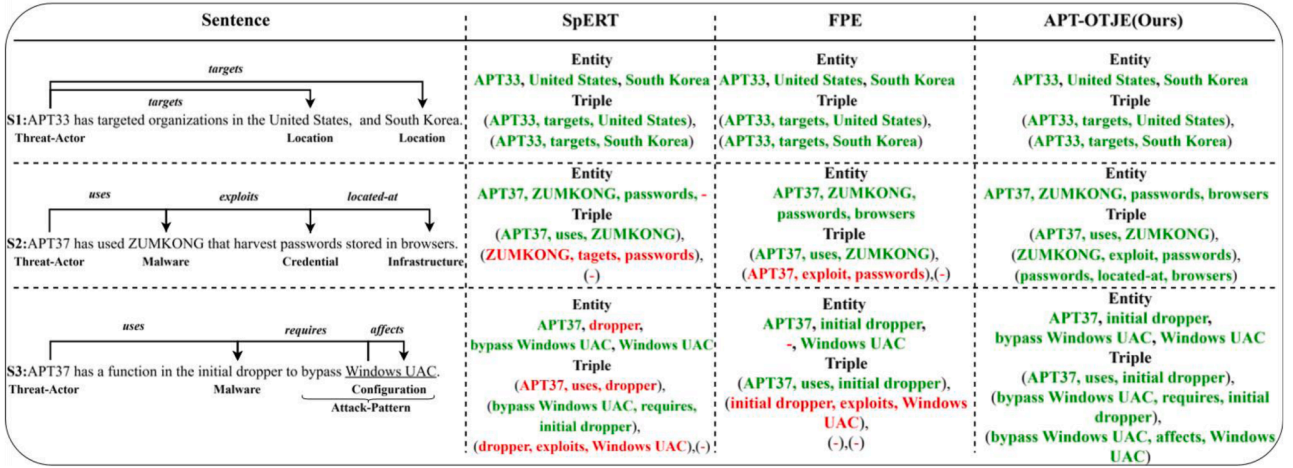


Fig. 4. Case study of overlapping triplets.

7. Robustness and limitations

Although this paper proposes an effective model for extracting overlapping triples in complex scenarios, the trained model still suffers from robustness and limitation issues, primarily from intrinsic data characteristics and extrinsic scale constraints.

The intrinsic characteristics of the data primarily cause robustness issues. While different overlapping triple types show relatively stable distribution patterns in unstructured APT texts, NEO types, despite being the most critical overlapping pattern, represent merely 6.4% of our dataset. When the three overlapping triple types are mixed during training, NEO types face unavoidable robustness challenges in training and real-world recognition.

Up-sampling-based data augmentation (Table 6) shows experimental improvements, this approach inevitably introduces bias. Because NEO-containing sentences inherently co-occur with SEO patterns, augmenting NEO samples simultaneously increases SEO instances, causing the model to disproportionately favor SEO recognition. This bias manifests as inflated SEO precision rather than balanced NEO improvement. Furthermore, same-type entity replacement may cause semantic inconsistency between substituted entities and their relational context, increasing the risk of false positives in relation classification. These factors collectively make up-sampling-based augmentation unreliable for deployment in scenarios where the triple type distribution differs from that of the training corpus.

Limitation issues stem from extrinsic scale constraints. Data imbalance across different triple types is the primary limitation of the current model. Although we filtered out numerous low-frequency triple types during training to reduce noise, the retained data still exhibits a long-tail distribution (ranging from 542 to 31 occurrences). Additionally, despite employing AI annotation tools during the team annotation process, cognitive biases among different review experts still introduce label inconsistency noise. This issue lurks beneath the data distribution imbalance and is easily overlooked.

The current model relies on SecureBERT and is therefore primarily applicable to English-language CTI texts. Extending the model to multilingual threat intelligence, where tokenization strategies, entity boundary conventions, and relation expression patterns differ substantially from English, remains an open challenge and is identified as a future research direction.

The case study in Section 6.5 provides structural-level qualitative insight by comparing the extraction outputs of APT-OTJE, SpERT, and FPE across representative sentences covering all three overlapping triple types. This comparison shows where and why each model succeeds or fails. For example, SpERT tends to misclassify relation types in long-

chain SOO structures, while FPE struggles with entity semantic enhancement in NEO scenarios. These observations offer indirect evidence of how the model responds to different structural configurations of overlapping triples. Token-level explainability, such as attention visualization or rationale extraction for individual relation predictions, is not addressed in the current work and remains a prioritized future research direction.

To address these challenges, future research will prioritize dataset expansion to mitigate triple type imbalance effects, followed by targeted model enhancements designed to improve robustness for overlapping triple characteristics.

8. Conclusion

This paper identifies three urgent problems in current joint entity-relation extraction in the APT domain: simple ontology structure definition, scarcity of publicly available datasets, and insufficient research on overlapping triple problems in complex scenarios. To address these issues, this paper first comprehensively defines four types of overlapping triples and focuses research in this domain on three types: SEO, SOO, and NEO. We then construct the CAPTIO ontology to address the inadequate definition of current joint extraction ontologies. Based on this ontology, we build and open-source our dataset CAPTIER. We propose the joint extraction model APT-OTJE to address APT overlapping triple problems. This model extracts overlapping triples using entity span-based approaches. It introduces the CTG mechanism in the feature extraction layer to actively represent shared information between tasks, improves positional encoding for entity and relation task characteristics, and learns different pattern features of multi-type triples through biafine multi-head mechanisms. Finally, we conduct comprehensive experimental validation of our method. APT-OTJE significantly outperforms baseline models in overall extraction performance, achieving improvements of 7% (CAPTIER dataset) and 25.2% (HACKER dataset). Performance on various overlapping triple extractions is intuitively validated through two datasets and case studies.

Data availability

Data that support the findings of this study are openly available in CAPTIER at <https://github.com/Bpig-C/CAPTIER>.

CRediT authorship contribution statement

Rui Qi: Writing – original draft, Validation, Software, Project administration, Methodology, Conceptualization. **Ga Xiang:** Writing –

review & editing, Methodology, Funding acquisition. **Lu Sun:** Investigation, Data curation. **Zicheng Tan:** Resources, Formal analysis. **Jun Cao:** Formal analysis. **Qunsheng Yang:** Data curation. **Mingyue Cheng:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the R&D Program of Beijing Municipal Education Commission (No. KM202311232014), the Curriculum Construction Project of Beijing Information Science and Technology University (No. 2024YKJ03), the General Program of Beijing Higher Education Association (No. MS2024277), and the XingGuang Program of Beijing Information Science and Technology University (No. XG2025ZD20).

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